

Image processing lab #8

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The screenshot displays a Google Colab environment. The main workspace contains two code cells. The first cell, labeled [1], imports necessary libraries (numpy, matplotlib, skimage, and numpy.fft) and loads a camera image, resizing it to (256, 256) pixels. The second cell, labeled [2], performs a 2D Fast Fourier Transform (FFT) on the image, applies a Laplacian filter, and visualizes the results using a 3x1 grid of subplots: the original image, the Laplacian spectrum (log magnitude), and the reconstructed image after filtering. To the right of the code editor is a Gemini chat window. It greets the user as 'Hello, leen' and offers assistance with Python libraries, Google Drive data loading, and ML model training. Below the chat is an 'Empty cell' for further code entry. The bottom status bar indicates the current time is 11:27 PM and the environment is Python 3.

```
[1] import numpy as np
import matplotlib.pyplot as plt
from skimage import data
from skimage.transform import resize
from numpy.fft import fft2, ifft2, ifftshift

image = data.camera()
image = resize(image, (256, 256))
M, N = image.shape

[2] u = np.fft.fftfreq(M).reshape(-1, 1)
v = np.fft.fftfreq(N).reshape(1, -1)
laplacian_filter = -4 * (np.pi ** 2) * (u**2 + v**2)

F = fft2(image)
F_lap = F * laplacian_filter
laplacian_image = ifft2(F_lap).real

plt.figure(figsize=(12, 5))
plt.subplot(1, 3, 1)
plt.imshow(image, cmap='gray')
plt.title('Original Cameraman')
plt.axis('off')

plt.subplot(1, 3, 2)
plt.imshow(np.log(1 + np.abs(F_lap)), cmap='gray')
plt.title('Laplacian Spectrum')
plt.axis('off')

plt.subplot(1, 3, 3)
plt.imshow(laplacian_image, cmap='gray')
plt.title('Reconstructed Image (Laplacian Filter)')
plt.axis('off')
```

Gemini

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[Load data from Google Drive](#)

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```
import numpy as np
from skimage import data
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from numpy.fft import fft2, ifft2, ifftshift

image = data.camera()
image = resize(image, (256, 256))
M, N = image.shape
```

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Variables Terminal 11:27 PM Python 3

lab 8

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Commands + Code + Text Run all

[2] ✓ 1s

```
plt.figure(figsize=(12, 5))
plt.subplot(1, 3, 1)
plt.imshow(image, cmap='gray')
plt.title('Original Cameraman')
plt.axis('off')

plt.subplot(1, 3, 2)
plt.imshow(np.log(1 + np.abs(F_lap)), cmap='gray')
plt.title('Laplacian Spectrum')
plt.axis('off')

plt.subplot(1, 3, 3)
plt.imshow(Laplacian_image, cmap='gray')
plt.title('Laplacian (Spatial Output)')
plt.axis('off')
plt.show()
```

Original Cameraman

Laplacian Spectrum

Laplacian (Spatial Output)

[3] ✓ 0s

```
def center_embed_kernel(kernel, shape):
    padded = np.zeros(shape)
```

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```
import matplotlib.pyplot as plt
from skimage import data
from skimage.transform import resize
from numpy.fft import fft2, fft, ifftshift

image = data.camera()
image = resize(image, (256, 256))
M, N = image.shape
```

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[3] ✓ 0s

```
G_y = ifft2(F * H_y).real
sobel_magnitude = np.sqrt(G_x**2 + G_y**2)

plt.figure(figsize=(15, 5))
plt.subplot(1, 3, 1)
plt.imshow(G_x, cmap='gray')
plt.title('Sobel X')
plt.axis('off')

plt.subplot(1, 3, 2)
plt.imshow(G_y, cmap='gray')
plt.title('Sobel Y')
plt.axis('off')

plt.subplot(1, 3, 3)
plt.imshow(sobel_magnitude, cmap='gray')
plt.title('Sobel Magnitude')
plt.axis('off')
plt.show()
```

Sobel X

Sobel Y

Sobel Magnitude

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image = resize(image, (256, 256))
M, N = image.shape
```

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